Magnet Monitor 4 (MM4)

Operating and Service Manual

5804162
Revision 2

Language Policy

DOC0371395 - Global Language Procedure

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. • Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. • Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. • Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 • 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤 害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 • 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 • 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 • 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他的危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 • 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 • 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 • 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. • Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. • Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. • Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.
VÝSTRAHA (CS)	Tento provozní návod existuje pouze v anglickém jazyce. • V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka. • Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah. • V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.
ADVARSEL (DA)	Denne servicemanual findes kun på engelsk. • Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse. • Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual. • Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAAR-SCHUWING (NL)	Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar. • Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan. • Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is. • Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.
WARNING (EN)	This service manual is available in English only. • If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. • Do not attempt to service the equipment unless this service manual has been consulted and is understood. • Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
HOIATUS (ET)	See teenindusjuhend on saadaval ainult inglise keeles. • Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest. • Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist. • Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.
VAROITUS (FI)	Tämä huolto-ohje on saatavilla vain englanniksi. • Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla. • Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen. • Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.
ATTENTION (FR)	Ce manuel d'installation et de maintenance est disponible uniquement en anglais. • Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire. • Ne pas tenter d'intervenir sur les équipements tant que ce manuel d'installation et de maintenance n'a pas été consulté et compris. • Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.
WARNUNG (DE)	Diese Serviceanleitung existiert nur in englischer Sprache. • Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen. • Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben. • Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙ ΗΣΗ (EL)	Τοπαρόν εγχειρίδιο σέρβις διατίθεται στα αγγλικά μόνο. • Εάν το άτομο παροχής σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει υπηρεσίες μετάφρασης. • Μηνεπιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό εκτός εάν έχετε συμβουλευτεί και έχετε κατανοήσει το παρόν εγχειρίδιο σέρβις. • Εάν δεν λάβετε υπόψη την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στο άτομο παροχής σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.
FIGYELMEZ- TETÉS (HU)	Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el. • Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése. • Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték. • Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.
AÐVÖRUN (IS)	Þessi þjónustuhandbók er aðeins fáanleg á ensku. • Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskip-tamanns að skaffa tungumálaþjónustu. • Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin. • Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.
AVVERTENZA (IT)	Il presente manuale di manutenzione è disponibile soltanto in lingua inglese. • Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione. • Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto. • Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manuten-zione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.
警告 (JA)	このサービスマニュアルには英語版しかありません。 • サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。 • このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。 • この警告に従わない場合、サービスを担当される方、操作員あるいは患者 さんが、感電や機械的又はその他の危険により負傷する可能性があります。
경고 (KO)	본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다. • 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다. • 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오. • 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.
BRĪDINĀJUMS (LV)	Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā. • Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodroši-nāt tulkojumu. • Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas. • Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.
ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării şi înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.

UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodilačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputstvo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja servisera, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNE- NIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obluhu a neporozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, obsluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohrozenie.
ATENCIÓN (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servicios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.
VARNING (SV)	Den här servicehandboken finns bara tillgänglig på engelska. • Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster. • Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandbo-ken. • Om du inte tar hänsyn till den här varningen kan det resultera i skador på serviceteknikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.
OPOZORILO (SL)	Ta servisni priročnik je na voljo samo v angleškem jeziku. • Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod. • Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli. • Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevar-nosti poškoduje ponudnik storitev, operater ali bolnik.
DİKKAT (TR)	Bu servis kılavuzunun sadece ingilizcesi mevcuttur. • Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer. • Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz. • Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.
ЗАСТЕРЕЖЕН НЯ (UK)	Даний посібник з експлуатації доступний тільки англійською мовою. • Якщо постачальник послуг клієнта спілкується іноземною мовою, тоді клієнт зобов'язаний забезпечити переклад. • Заборонено проводити огляд обладнання без попереднього звертання до даного посібника з експлуатації і розуміння інформації, поданої у ньому. • Недотримання цього застереження може завдати шкоди здоров'ю постачальника послуг, оператора або пацієнта через ураження електричним струмом, механічну травму або інше ушкодження.

Contents

1

Chapter 1 Prerequisites

Safety

Safety
Before working in any GE Healthcare MR suite or performing any GE Healthcare service procedure, you must: • Have read and understood all hazard conditions and safety requirements in the latest revision of the GE Healthcare *MR Service Safety Manual* (5452735). • Have successfully completed all relevant GE Healthcare Environmental Health and Safety (EHS) courses (or for non-GE employees, equivalent workplace training courses). • Comply with all site-specific training and workplace safety requirements. If you have any safety concerns at any time, do not begin work or immediately stop work and move to a safe location. Immediately contact your supervisor or site safety officer for instructions on how to proceed.

The most current version of the *MR Service Safety Manual* (5452735) should be available at all times while completing any actions contained within this document. One should make sure that they are working with the most current (latest release) of the *MR Service Safety Manual* before and while completing any actions contained within this document.

The latest release of the *MR Service Safety Manual* can be gotten through the support documentation library or through your GE Healthcare Field Service Representative.

Training and knowledge required for operator

No training is required for operator.

Training and knowledge required for service provider

GE MR service training is required for the service provider who will install, configure, and use the data on the Magnet Monitor. The service provider must understand the basic cryogenic system.

Environmental restrictions on locations of use

-
- Mounting location: The Magnet Monitor unit is a wall mount, or on either side of system equipment cabinet. The unit should not be mounted in the exhaust path of any equipment cabinet. The Magnet Monitor's power cord length is 1829 mm (72 inches). A power receptacle must be accessible within the length of the power cord. See the Electrical Specifications table on page 11 for more detailed requirements.
- Magnetic field: The maximum gauss limit for the Magnet Monitor is 200 gauss (20 mT). The unit should not be mounted in a magnetic field greater than this limit.
- Distance to other electronic devices (cell phones, and so on): The Magnet Monitor is installed in the MR equipment room or operating room. Refer to the *System Pre-Installation Manual* for guidelines regarding placement.

2

Chapter 2 Specifications

2.1 Magnet Monitor component

Replacement parts		
Item	**Quantity**	**Part number**
Magnet Monitor 4	1	5771373

The Magnet Monitor is a stand-alone computer-based device that interfaces to various types of magnet, cryocooler, and system cabinets to provide sensor loop control, analog and digital data acquisition, alarm setting and handling, as well as means for remote monitoring and diagnostics.

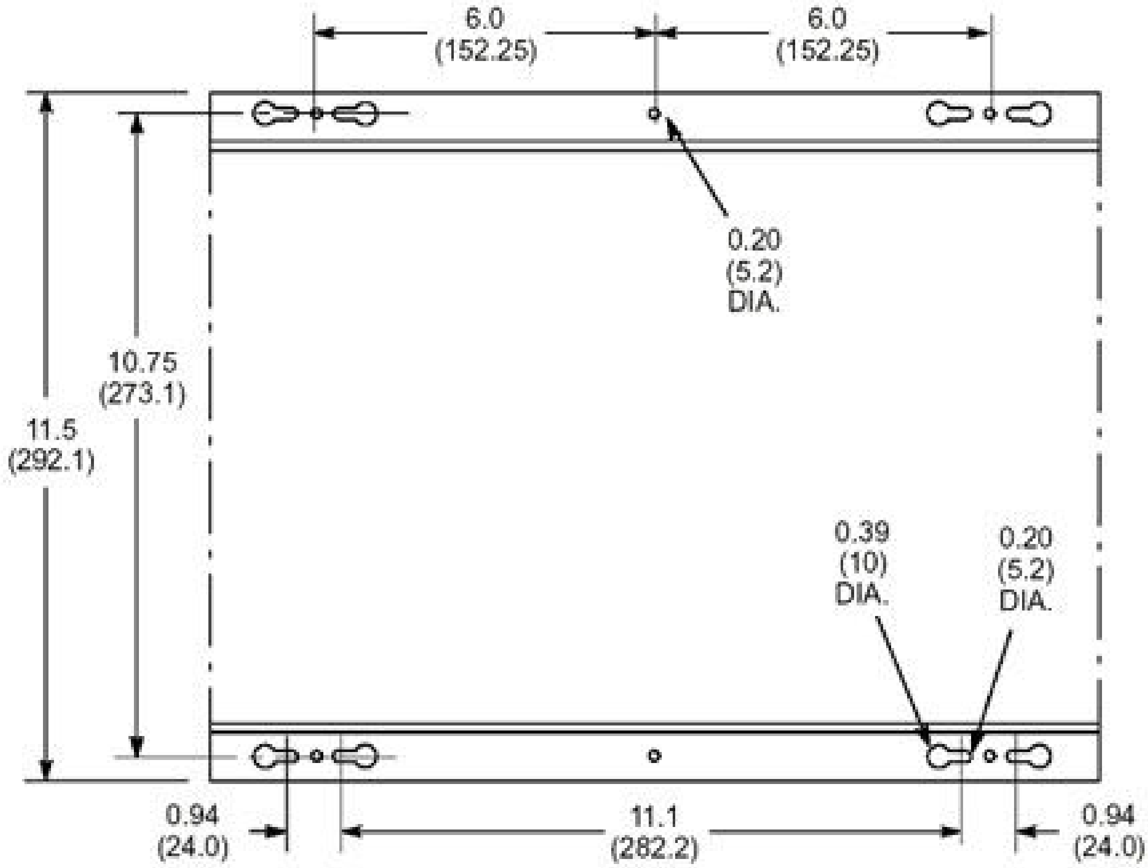
It constantly monitors the state of the superconducting magnet through temperature, pressure, and liquid helium level sensors. The Magnet Monitor uses a heater to maintain the correct pressure in the magnet's helium vessel. It constantly monitors the health of the cryogenic cooling system and resets the compressor if the unit trips under certain conditions. The Magnet Monitor reads a signal from the system cabinet and uses this signal to filter out noises in data caused by system scanning.

The Magnet Monitor provides two interfaces with which the user can interact:

- Front panel keypad, LCD display, and LED lights
- Web interface

Mechanical specifications	
Parameter	**Specification**
Dimensions	• Height: 254 mm (10.0 inches) • Width: 381 mm (15.0 inches) • Depth: 76 mm (3.0 inches)
Weight	3.6 kg (8 lbs)
Mounting	Wall mount; for systems with Heat Exchange Cabinet (HEC) or Integrated Cooling Cabinet (ICC), this unit mounts on either side of the cabinet.
Service clearance	Maintain 304.8 mm (12 inch) clearance above, below, and on both sides of the unit.

Figure 2-1 Magnet Monitor dimensions



NOTE

All dimensions in diagram are in inches. All bracketed () dimensions are in millimeters.

Environmental specifications	
Parameter	**Specification**
• Operating temperature • Nonoperating temperature	• 0 to 40°C (32 to 104°F) • -34 to 50°C (-29.2 to 122°F)
Temperature change	5°C/h
• Operating relative humidity • Nonoperating relative humidity	• 10 to 80% (noncondensing) • 10 to 90% (noncondensing)
Magnetic field	< 200 gauss (20 mT)
Altitude	-30 to 2600m, above sea level
Atmospheric pressure	70 to 105 kPa

Electrical specifications	
Parameter	**Specification**
Input voltage	100 to 120 VAC or 200 to 220 VAC (auto switching)
Line frequency	50 or 60 Hz (auto switching)
Nominal current	0.5A
Power outlet rating	2.0A

NOTE

The Magnet Monitor requires a 110/220 VAC, 50/60 Hz, 2.0A facility-supplied outlet. Power at the outlet must be continuously available.

NOTE

The Magnet Monitor's power cord length is 1829 mm (72 inches). The power outlet must be accessible within the length of the Magnet Monitor's power cord and not blocked by other objects.

Classification	
Class	**Specification**
Class I, IPX0 (not protected), and Continuous Operation	WARNING - CLASS 1 ME EQUIPMENT – MUST BE CON-NECTED TO GROUNDED AC MAIN RECEPTACLE

2.2 RUO preamplifier

Replacement parts		
Item	**Quantity**	**Part number**
RUO Preamplifier	1	2219341

Figure 2-2 RUO preamplifier

Mechanical specifications	
Parameter	**Specification**
Dimensions	• Width: 50.8 mm (2 inches) • Length: 95.25 mm (3.75 inches) • Depth: 38.1 mm (1.5 inches)
Weight	142g (5 oz)
Mounting	• Surface mount with Velcro • On top of the magnet near the coldhead

Environmental specifications	
Parameter	**Specification**
• Operating temperature • Nonoperating temperature	• 0 to 40°C (32 to 104°F) • -34 to 50°C (-29.2 to 122°F)
Temperature change	5°C/h
• Operating relative humidity • Nonoperating relative humidity	• 10 to 80% (noncondensing) • 10 to 90% (noncondensing)
Noise	< 50 dB
Magnetic field	< 1500 gauss
Altitude	-30 to 2133m, above sea level

Electrical specifications	
Parameter	**Specification**
Input voltage	± 12 VDC
Gain per channel	65 ± 1%
Channel to channel isolation	≥ 60 dB
Output noise	≤ 50 µV rms
Bandwidth	3 dB bandwidth > 10 kHz

Input connector J1; to/from sensors; type DB15; female		
Pin	**Signal description**	**Signal level**
1	Stage 1 diode positive	10 µA constant current
2	Stage 1 diode negative	
3	Stage 2 diode positive	10 µA constant current
4	Stage 2 diode negative	
5	RUO (1) source positive	10 µA constant current
6	RUO (1) source negative	
7	RUO (1) signal positive	0 to ± 20 mV DC
8	RUO (1) signal negative	
9	No connection	N/A
10	No connection	N/A
11	No connection	N/A
12	RUO (2) source positive	10 µA constant current
13	RUO (2) source negative	
14	RUO (2) signal positive	0 to ± 20 mV DC
15	RUO (2) signal negative	

Output connector J2; to/from Magnet Monitor; type DB15; male		
Pin	**Signal description**	**Signal level**
1	Stage 1 diode positive	10 µA constant current

Output connector J2; to/from Magnet Monitor; type DB15; male		
Pin	**Signal description**	**Signal level**
2	Stage 1 diode negative	
3	Stage 2 diode positive	10 µA constant current
4	Stage 2 diode negative	
5	RUO (1) source positive	10 µA constant current
6	RUO (1) source negative	
7	RUO (1) output	0 to ± 2.5 VDC
8	Signal ground	0 VDC
9	No connection	N/A
10	No connection	N/A
11	No connection	N/A
12	RUO (2) source positive	10 µA constant current
13	RUO (2) source negative	
14	RUO (2) output	0 to ± 2.5 VDC
15	RUO (2) signal negative	0 VDC

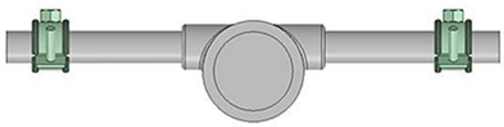
2.3 Water flow/temperature meter (turbine type)

Replacement parts		
Item	**Quantity**	**Part number**
Water Flow/Temperature Meter (Turbine Type)	1	2333825

Mechanical specifications	
Parameter	**Specification**
Dimensions	• Height: 69.9 mm (2.75 inches) • Width: 94 mm (3.70 inches) • Depth: 50.8 mm (2.0 inches)
Weight	1.3 kg (2.75 lbs)
Mounting	Horizontal with water pipes on the top on any vertical surface. See Figure 2-3 Flow meter mounting orientation on page 15 below. **NOTE** Must not lay on floor; this will result in performance issues.
Maximum fluid pressure	200 psig
Maximum fluid temperature	107.2°C (225°F)
Inlet/outlet piping	1/2 inch NPT x 5-inch long brass pipe. **NOTE** Straight pipe required for non-turbulent laminar flow.

Mechanical specifications	
Parameter	**Specification**
Non-wetted materials composition	• Epoxy • Lexan • PVC
Wetted materials composition	• 316 stainless steel • Buna-N • Delrin • Acetal Copolymer
Flow direction	Operates with flow from either direction.

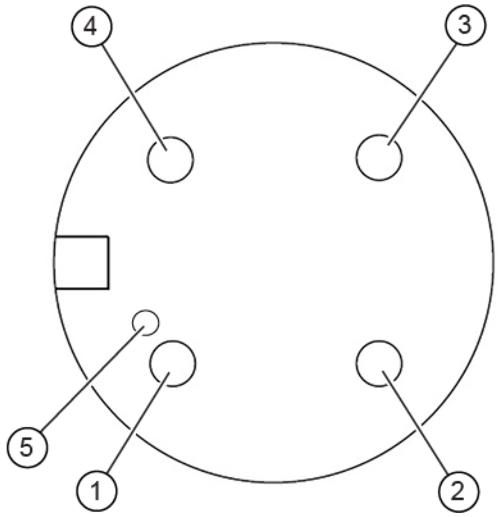
Figure 2-3 Flow meter mounting orientation



Environmental specifications	
Parameter	**Specification**
• Operating temperature • Nonoperating temperature	• 0 to 40°C (32 to 104°F) • -34 to 50°C (-29.2 to 122°F)
Temperature change	5°C/h
• Operating relative humidity • Nonoperating relative humidity	• 10 to 80% (noncondensing) • 10 to 90% (noncondensing)
Magnetic field	< 10 gauss

Electrical specifications	
Parameter	**Specification**
Input voltage	+12 to 35 VDC, < 25 mA
Transmission distance	< 6096 cm (200 ft)
Output voltage (analog)	0 to 5 VDC, for both flow and temperature
Calibrated flow range	0 to 5 gpm (0 to 18.9 lpm)
Calibrated temperature range	-18 to 38°C (0 to 100°F)
Accuracy	± 2% of range
Repeatability	0.5% of range
Resolution	Infinite
Response time (flow time)	2s to 90% step change
Calibration	Factory-calibrated, no service calibrations required

Figure 2-4 Electrical connections - wiring side



Pin 1	+ Positive DC supply
Pin 2	Common ground
Pin 3	Flow sensor analog output
Pin 4	Temperature sensor analog output
5	Raised dimple

3

Chapter 3 Product structure

The Magnet Monitor is packaged in a collector. Depending upon the collector ordered with your system it may also ship with the RUO preamplifier box or the water flow meter. A serial cable, 5807125, for connecting with the F50SH compressor may also be included. This additional cable will connect the RS232 port on the Magnet Monitor (J2) to RS232 port on the F50SH.

If any parts are found to be missing, submit a Missing In Shipment report to receive parts. Information can be found in the front material of the installation manual.

If any received part is found to be defective, then code as DOA and order a replacement.

Refer to the respective *System Installation Manual* or Service Methods for exact configuration.

4

Chapter 4 Installation and upgrades

4.1 Magnet Monitor installation

A GE field service engineer or qualified service provider is responsible for installing and configuring the Magnet Monitor. This chapter provides instruction on how to install and configure a Magnet Monitor unit.

4.1.1 Features

The Magnet Monitor consists of the front panel display, user keypad hardware, and software used for the following:

- Measure helium vessel level
- Measure helium vessel pressure
- Provide helium vessel pressure control on magnets that have recondenser systems
- Measure coldhead and magnet shield/recondenser temperatures
- Measure pressure control heater duty cycle
- Measure water flow and temperature for the water-cooled compressor
- Monitor status of compressor
- Auto restart the compressor after interruption in power or cooling

4.1.2 Compatibility

Magnet compatibility

The Magnet Monitor is designed to interface with the following GE-manufactured magnets, as well as other magnet with similar configurations:

- LCC or Rxxx, RAxxx, Qxxxx, CRxxxx
- LCC300 or Wxxxx, WBxxxx
- LCC_HM (1.5T Wide) or HMxxx
- LCC_UA (3T Wide) or UAxxxx
- LCC_W or RDxxxx
- Platform 1.5T or PMxxxx or PCxxxx
- Ares 3.0T Magnet ARxxxx or ACxxxx

Shield cooler/cryocooler compressor compatibility

The Magnet Monitor is designed to interface with the following compressors:

- Sumitomo models: CSW-71, CSA-71, CNA-31, CNA-61, F-50, FA-50SH/L, F50SH, F70, and HC10

NOTE

A water flow/temperature sensor is externally installed in the coolant supply line except for air cooled compressors.

4.1.3 Installing and setting up sequence

About this task

Use the recommended sequence of steps for installing a new Magnet Monitor.

Procedure

1. Review inventory of all hardware according to the appropriate collector for your system before installation.
2. Install the Magnet Monitor unit. See 4.3 Installing hardware on page 21.
3. Install and connect the Magnet Monitor system cables. Refer to the appropriate cable kit and wiring diagram for your system. Different MR systems require different Magnet Monitor cable kits and may have different interconnect maps.
4. Apply AC power to the Magnet Monitor. See the Electrical specifications table on page 11.
5. After the boot-up is complete, the Magnet Monitor unit's LCD display will show its software revision number. Connect a PC to the Magnet Monitor's Ethernet.
   - See 7.1 Connecting through the building network on page 29, or use direct connect if the site LAN does not have DHCP.
   - See 7.2 Connecting a direct network cable to a laptop and the Magnet Monitor using Windows 10 on page 30.
6. Log on to the Magnet Monitor's Web interface. See Chapter 8 Logging on to the Magnet Monitor as the customer on page 36 and 9.1 Basic web page layout on page 39.
7. Set the date and time on the Magnet Monitor. See 9.2.1 Setting the date and time on page 39.
8. Select the correct magnet type and compressor type. See 9.2.2 Setting the magnet type on page 40.
9. Enter the hospital and location data, magnet serial number, and site elevation.
10. Establish the correct network settings on the Network page. See 9.2.3 Setting the network IP configuration on page 45.
11. Click **Logout** on the top left side of the web page to activate the new settings.

4.2 Tools and test equipment

Tools and test equipment		
Item	**Quantity**	**Part number**
Nonmagnetic Tool Kit (Basic)	1	2385098
Nonmagnetic Titanium Service Tool Kit, Small Set	1	5112581 or 5113258 for 3T and 1.5T
Nonmagnetic Tool Kit, Inch/Metric		46-320273G4 for 1.5T only

Tools and test equipment		
Item	**Quantity**	**Part number**
Nonferromagnetic Tape Measure, 1.828m (6 ft) Long, Minimum or Non-ferrous Metric Ruler	1	-
Adjustable Wrench, 8 inch, Titanium	1	5109896-2
Nonmagnetic 9/16 inch Combination Wrench	1	-
Flat-Blade Screwdriver, Small	1	-
Nonferrous Metric Ruler	1	-
Utility Knife	1	-
Nut Driver, 3/16 inch	1	-
Nut Driver, 1/4 inch	1	-
Digital Voltmeter (DVM)	1	46-194427P284
Phillips #2 Screwdriver	1	-
Wrist Grounding Strap	1	46-198094P1
Shallow Container (to catch water after disconnecting the hose to the Shield Cooler Compressor)	1	-
Tools and hardware to mount Magnet Monitor to wall: • Power Drill • Drill Bits • Wall Anchor System	1 of each	-
Laptop Computer	1	-
Crossover Network Cable	1	2394952
M4-0.8 X 20 mm Phillips Pan Head Screw for mounting to Heat Exchange Cabinet (HEC), not included	6	-
Safety Glasses	1 pair	-
Safe Skin Nitrile Glove, Large	1 pair	2207303-6
Pair: Nonferrous Safety Shoes	1 pair	-
Pair: Cut-Resistant Gloves	1 pair	-

4.3 Installing hardware

4.3.1 Magnet Monitor wall installation

For MR450, MR450w, MR750, MR750w, Artist, Architect, and PET/MR systems, the Magnet Monitor should be attached to the heat exchange cabinet (HEC). For other systems, the Magnet Monitor unit can be mounted on a wall surface that is not blocked by other objects and can be easily accessed by users. Mounting orientation should be same as the illustration below so the user can read information on the LCD display with no difficulty. Refer to the *System Installation Manual* and site equipment location drawings for more details.

- For installed base systems receiving a Magnet Monitor for the first time, make sure the location complies with the equipment specifications for gauss levels and service clearance. See 2.1 Magnet Monitor component on page 10.
- For installed base systems that already have MM3 and are upgrading to MM4, the hole pattern is the same.
- For installed base systems that already have an MM2 (SHeM manufactured by Granite) and are upgrading to a Magnet Monitor, mark new hole locations on the wall because the mounting hole patterns are different. For mounting hole dimensions, see 2.1 Magnet Monitor component on page 10.

Figure 4-1 Magnet Monitor mounting on HEC

- Note the SSH account's password information on the label, located at the back side of Magnet Monitor, before mounting it.

Anchor the Magnet Monitor to the wall using all applicable codes for the site. Make sure a 24 x 7 non-switched 120/240 VAC power source is available within 2m (6 ft) of the mounting location of the Magnet Monitor. The Magnet Monitor automatically adjusts to line voltage and line frequency, so no further adjustments are required.

- Do not connect the AC power cord to the Magnet Monitor until all cabling is completed (the Magnet Monitor does not have an ON-OFF switch for power control).
- If preferred, the Magnet Monitor may be plugged into a 220 VAC outlet. Get a power cord with the appropriate connector.

4.3.2 Magnet room cable connections (all systems)

Refer to the appropriate cabling configuration for your system, and the respective *System Installation Manual* or Service Methods for the exact interconnect map.

4.3.3 Equipment room cable connections (all systems)

Refer to the appropriate cabling configuration for your system, and the respective *System Installation Manual* or Service Methods for the exact interconnect map.

4.3.4 Unit power on and off

There is no power On/Off switch in the unit since the intended operation is to be on continuously.

- To power On the unit, plug Magnet Monitor's AC main power cord into the electricity outlet.
- To power Off the unit, unplug Magnet Monitor's AC main power cord from the electricity outlet.

4.4 Calibration and maintenance

No calibration is required. For a guideline of routine inspection and maintenance, refer to the *MR System Planned Maintenance* schedule and procedures.

5

Chapter 5 Interconnect maps

Refer to the respective *System Installation Manual* or Service Methods for the exact interconnect map.

Chapter 6 Front panel user interface

6.1 Introduction

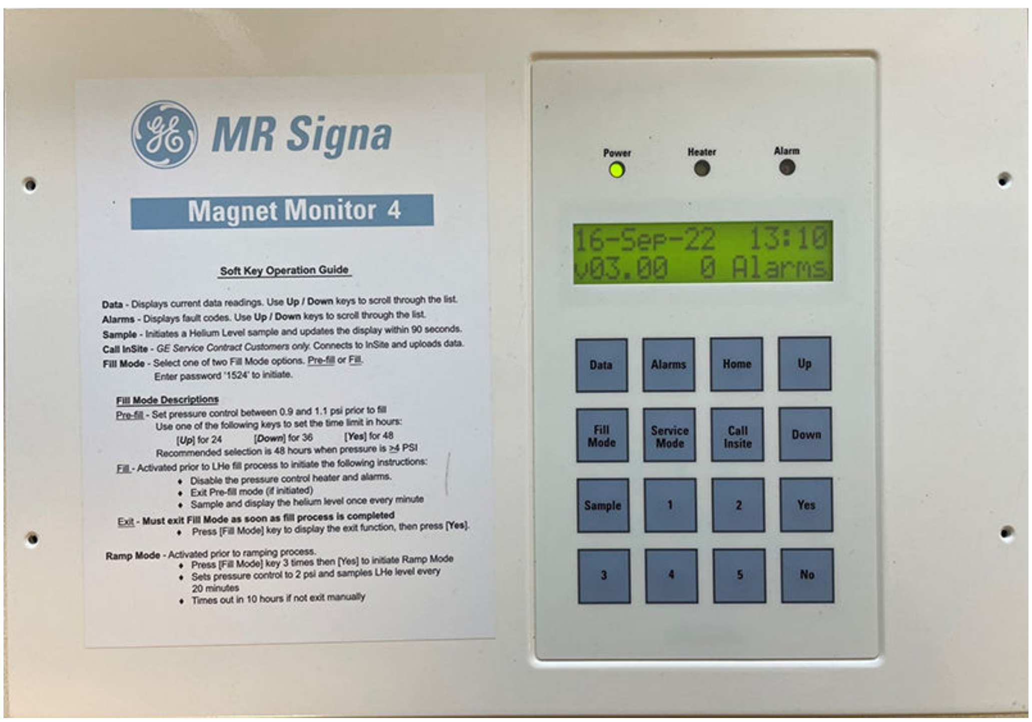
The front panel of the Magnet Monitor unit is made up of the following elements:

- LEDs showing AC power, heater activity, and alarm activity.
- LCD display for user interaction.
- 16-button softkey pad.

This LCD display is controlled by the Magnet Monitor's CPU, and provides easy viewing of the date/time, real-time sensor data, and fault codes. Other features include the setting of Fill modes and Service modes through predefined access codes.

The flexible membrane keypad contains 16 buttons. For instruction on their operation, see 6.2 Using the Magnet Monitor front panel on page 25.

Figure 6-1 Magnet Monitor front panel user interface



Power LED	(Green) Lit when the Magnet Monitor is powered on
Heater LED	(Yellow) Lit when the heater is on
Alarm LED	(Red) Lit when there is an alarm condition

6.2 Using the Magnet Monitor front panel

6.2.1 Softkey buttons

Data

Pressing **Data** causes the Magnet Monitor unit to show a list of sensor parameters and real-time data. Press **Up** and **Down** to scroll through the list of items. A single item can be shown indefinitely and the Magnet Monitor will update that value (except for the He level) one time per minute.

NOTE

Leaving the He level parameter on the page will not force the Magnet Monitor to sample a new He level value each minute. You must enter Fill mode to generate a new helium level sample each minute or press **Sample** to update the reading one time.

NOTE

Cycling through the Data button entries will show the current IP address and InSite2 connection status.

Alarms

Pressing **Alarms** causes the Magnet Monitor to show a list of current fault codes. Press **Up** and **Down** to scroll through the list of items. You can only update items in this list by exiting from the Alarm menu and then reentering the menu. The alarm LED is lit if there are any fault codes to view.

NOTE

The Magnet Monitor tracks potential faults over a 10-minute period before the software activates the alarm LED and updates the fault codes to minimize false reporting due to noisy transients.

Home

Pressing **Home** puts the display into normal operating mode. In this mode, the display will cycle through the following information.

NOTE

When you complete another operation, it is best to return to the Home display.

Table 6-1 Home display

Page	Display	
1	Date	Time
	Software revision	Number of alarms
2	He level (percentage or liters)	
	`x.xx%`	
3	He pressure	
	`x.xxx psi`	

Call Insite

This function is not used in non-proprietary mode.

Sample

Pressing **Sample** causes the Magnet Monitor to initiate a helium level sample and read the current helium level. It may take up to 90 seconds to update the display for the new level(s).

6.2.2 Fill mode button

Pressing **Fill mode** lets you select one of four options: Pre-Fill mode, Fill mode, Ramp mode, or Storage mode. The code for all Fill mode selections is `1524`.

Pre-Fill mode

Pre-Fill mode is used for LCC-type magnets before a helium fill. This option will change the pressure control settings that are used for heater cycling. The end result is the magnet's helium vessel pressure is reduced from the normal setting down to approximately 1.0 psig as long as the cryogen cooler cabinet coldhead and recondenser are functioning normally.

NOTE

Pre-Fill mode will run for up to 7 days.

If the user activates Fill mode before this time period expires, Pre-Fill mode is automatically exited and the Fill mode functions take over.

If the time period expires or you intentionally exit Pre-Fill mode using the soft keypad, the software will automatically put heater cycling into a post-fill operation. During post-fill, the operating parameters for pressure control are changed and the heater will be activated for an extended period of time, letting the magnet's helium vessel pressure return to its normal operating state. When in post-fill, heater cycling will not activate any heater alarms for 48 hours.

NOTE

Pre-Fill mode should only be used for LCC, LCC300, LCC_W, LCC_HM, LCC_UA, Platform 1.5T, and Ares 3.0T magnet types.

Fill mode

Fill mode does the following functions:

- Exits Pre-Fill mode (if necessary).
- Disables the pressure control function for heater cycling.
- Initiates a helium level sample one time every minute and refreshes the LCD display information.

You must select and initiate Fill mode before inserting the fill line stinger into the magnet. When the filling operation is complete, you will exit the Fill mode option and the software will automatically go into a post-fill operation. During post-fill, the operating parameters for pressure control are changed and the heater will be activated for an extended period of time, letting the magnet's helium vessel pressure return to its normal operating state. When in post-fill, heater cycling will not activate any heater alarms for 48 hours.

Ramp mode

Ramp mode does the following functions:

- Sets the Magnet Monitor to Ramp mode.
- Shows the Ramp mode message on the front display.
- Holds the magnet pressure at 2.0 psig and samples helium level every 20 minutes.
- Ramp mode times out after 10 hours.

You must select and initiate Ramp mode before inserting the ramp probes into the magnet. When the magnet ramping is completed, exit Ramp mode option, putting the Magnet Monitor into post-fill operation. During post-fill, the operating parameters for pressure control are changed and the heater will be activated for an extended period of time, letting the magnet's helium vessel pressure return to its normal operating state. When in post-fill, heater cycling will not activate any heater alarms for 48 hours.

Storage mode

Storage mode does the following functions:

- Sets the Magnet Monitor to Storage mode.
- Holds the magnet pressure at 0.8 psi.
- Storage mode times out on reboot or after 2 weeks.

Storage mode can be activated using the keyboard if the magnet needs to be stored at a temporary location. The Magnet Monitor will keep the magnet pressure set point at 0.8 psi (digital gauge) to prevent unnecessary venting even if the magnet is accidentally left in transport configuration. If the magnet needs to be stored longer than 2 weeks, Storage mode must be reactivated.

6.2.3 Service mode button

Pressing **Service mode** lets you select one of these options:

- Web mode Same IP, using either DHCP or static IP address, assigned to unit by site IT
- Web mode Fixed IP, using fixed IP address (192.168.0.1)
- Service mode (for GE Service only)
- Reset passwords

Web mode

You must put the Magnet Monitor into Web mode to use the configuration tool's web interface. Any user, whether a GE employee or not, can connect and log on to the interface with a laptop or the system host computer. After you are logged on, the configuration settings of the Magnet Monitor can be changed and saved.

When you press **Service mode**, you must know what form of Web mode to enter. Pressing **Service mode** lets you select one of these options:

- Web mode - Same IP: uses either DHCP or static IP address, assigned to unit by site IT.
- Web mode - Fixed IP: uses fixed IP address (192.168.0.1).

Press **Yes** when the correct option is shown to put the Magnet Monitor in Web mode. No password entry is required.

NOTE

> The unit will remain in Web mode for 45 minutes, after which time it will terminate the connection. Web mode will also be terminated if the AC power to the unit is cycled off, then on.

When you are in Web mode, press **Data** repeatedly until the IP address shows. Write the address down, since you will need it when setting up your laptop or host computer to connect to the unit.

When you are in Web mode, you can now attempt to connect to the unit. See 7.1 Connecting through the building network on page 29 or 7.2 Connecting a direct network cable to a laptop and the Magnet Monitor using Windows 10 on page 30.

Service mode

This mode of operation is only accessible by GE field service personnel. Pressing **Service mode** one time will show the **Enter Service Mode? (Yes/No)** prompt. Press **Yes** to continue in Service mode.

The Magnet Monitor unit must be in Service mode before doing any of the following services, to prevent the unit from calling out and having the Online Center generate Request for Service (RFS) emails for this site:

- Coldhead, gas line, or compressor servicing
- Chiller servicing
- Any troubleshooting or repair to the magnet that would impact the helium vessel pressure or helium level (for example, shim lead replacement, pressure transducer replacement, and so on)

When the service session is complete, press **Service mode** to exit Service mode. The display should show **Exit Service Mode? (Yes/No)**. Press **Yes** to return to the Home display.

NOTE

If Service mode is not exited manually, the unit will remain in Service mode for 8 hours, after which time it will terminate the connection and return to the Home display. Service mode will also be terminated if the AC power to the unit is cycled off, then on.

Password reset

When activated, Password reset will reset all passwords used for the Web interface to default passwords. The default passwords can be changed to the desired passwords within 30 minutes. All default passwords that are unchanged within 30 minutes will be replaced with random strings. Back office information will be updated accordingly. The passwords for SSH accounts will not change.

7

Chapter 7 Laptop connection

7.1 Connecting through the building network

Prerequisites

Required conditions
The Magnet Monitor was previously configured for network operation using a direct connection between this device and a laptop.
The Magnet Monitor has been successfully communicating over the building network.
The Magnet Monitor is connected to the building network through the Magnet Monitor - J5 port.

About this task

This procedure is used to log on to the Magnet Monitor from the service browser on the customer's scanner or from a laptop attached to the customer's network.

Procedure

1. Find the IP address assigned to the Magnet Monitor by pressing **Data**, and then **Up** repeatedly until the IP address shows.

   NOTE

   After the IP is assigned by DHCP, the address will show correctly on the LCD instead of showing `Negotiating`.

2. Write down the IP address for use later.
3. Press **Service Mode** repeatedly until the display shows **Start Web Mode? (Yes/No) Same IP**.
4. Press **Yes** to activate Web mode.
5. Connect the laptop to the building network.
6. Open a web browser session on the laptop.
7. Type the Magnet Monitor's IP address (recorded earlier) into the laptop browser's address bar and click **Go** or press **Enter**.

   NOTE

   Some browsers may show a non-secure connection warning. This is normal and the connection will be confirmed.

   The browser should show the *Magmon4 Service Login* page.

8. Log on to the Magnet Monitor. See Chapter 8 Logging on to the Magnet Monitor as the customer on page 36.

7.2 Connecting a direct network cable to a laptop and the Magnet Monitor using Windows 10

About this task

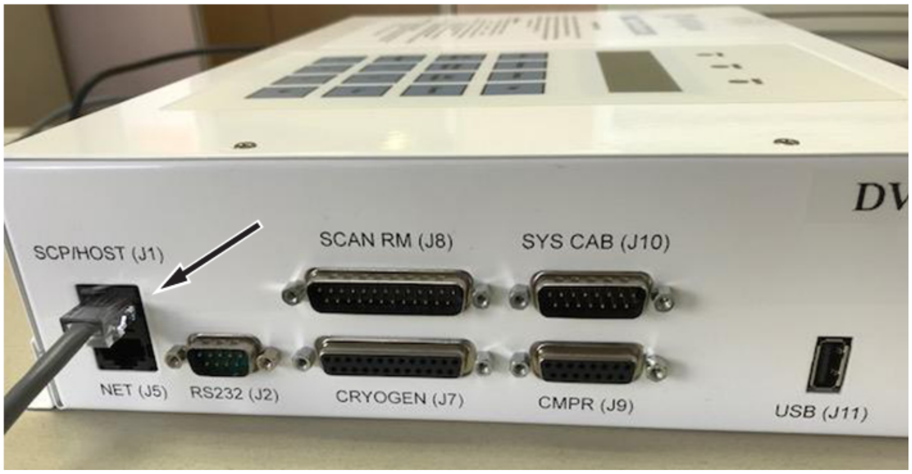
The Magnet Monitor provides the following Ethernet ports:

- J5 (Net) for connectivity with the GE back office.
- J1 (SCP/Host) for future expansion.

NOTE

To avoid interruption of connectivity with the back office, you should always use SCP/HOST (J1) for connection with the service laptop or PC when using the direct connection. When using a direct connection to the Magnet Monitor, make sure to use the appropriate network cable.

Figure 7-1 SCP/HOST (J1) connection

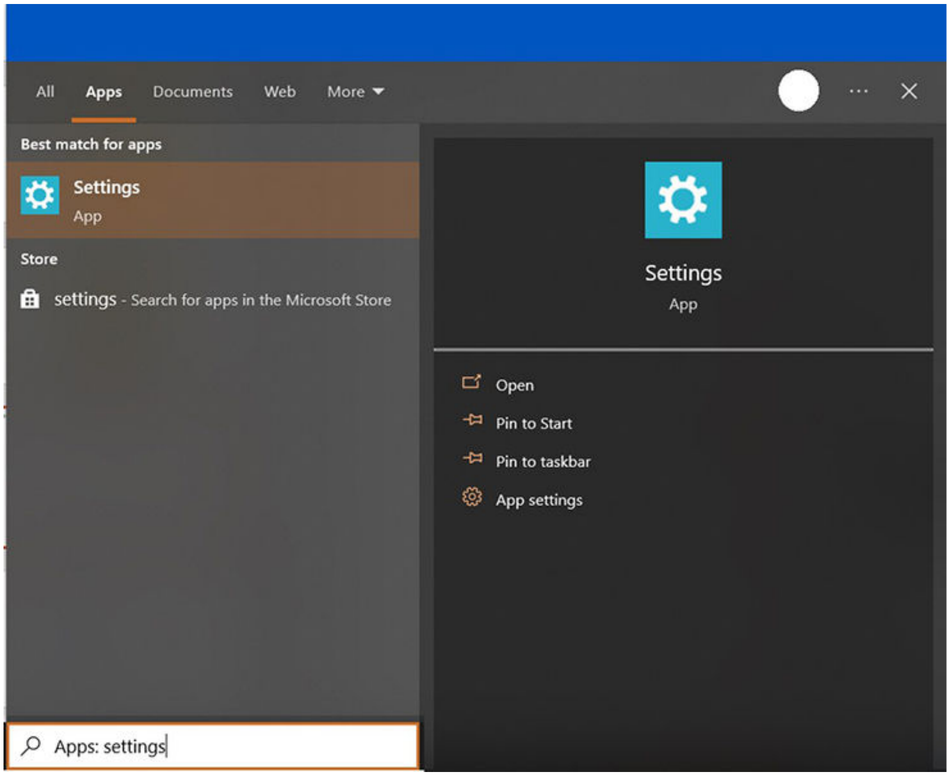


Procedure

1. Click the **Start** menu on the laptop and enter `Settings`.

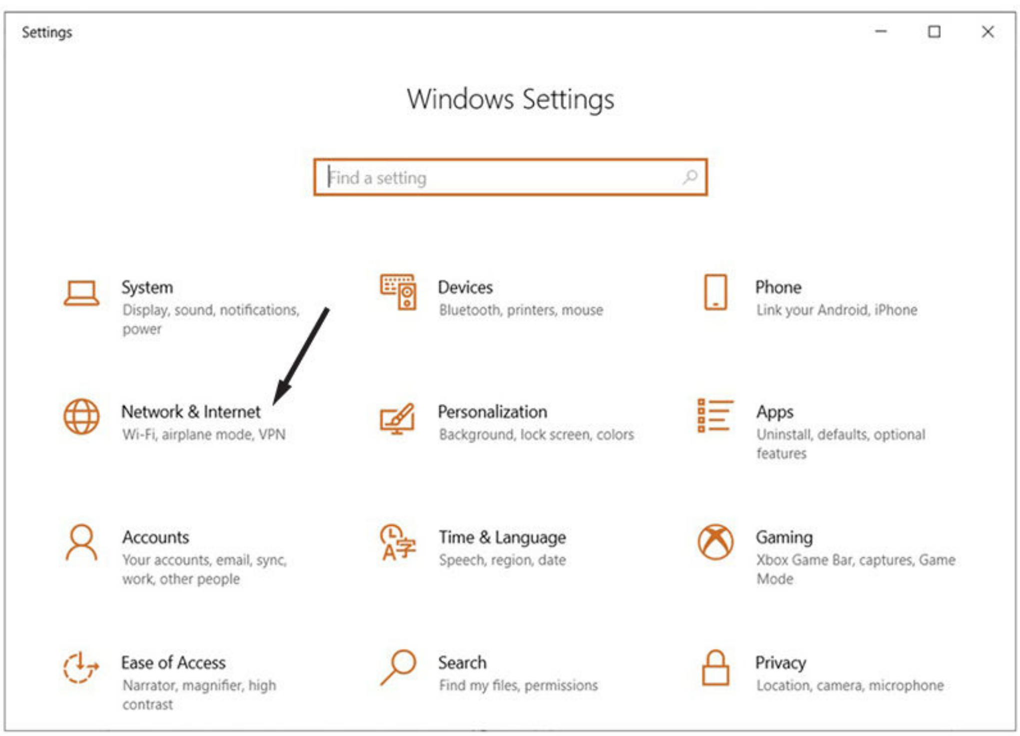
2. Click the **Settings App** option.

Figure 7-2 Settings App option



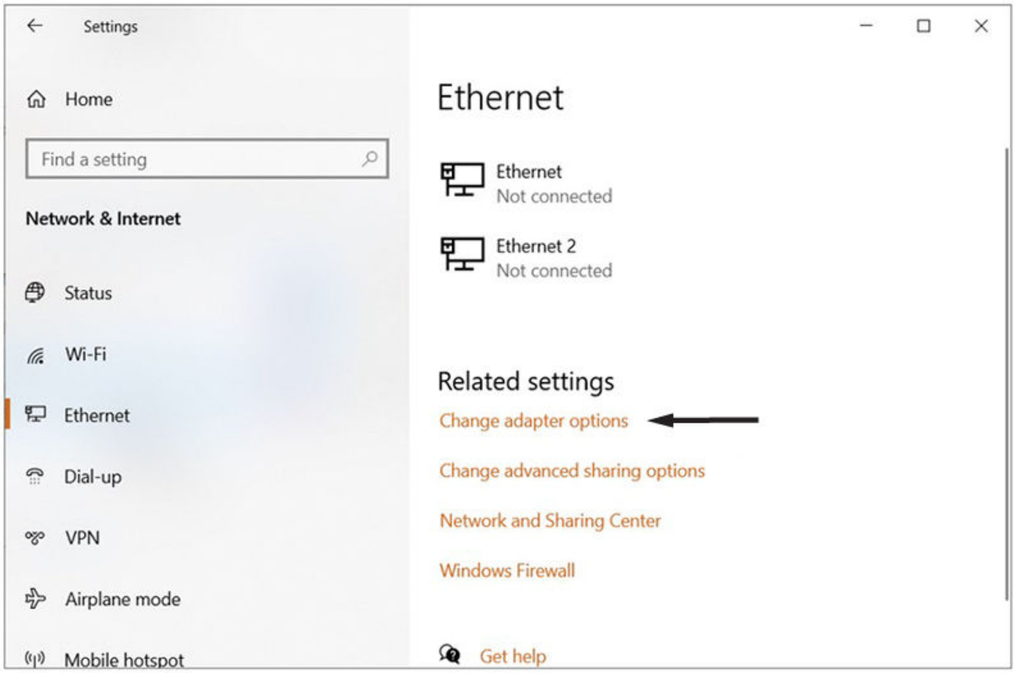
3. On the *Windows Settings* page, click **Network & Internet**.

Figure 7-3 Windows Settings - Network & Internet option



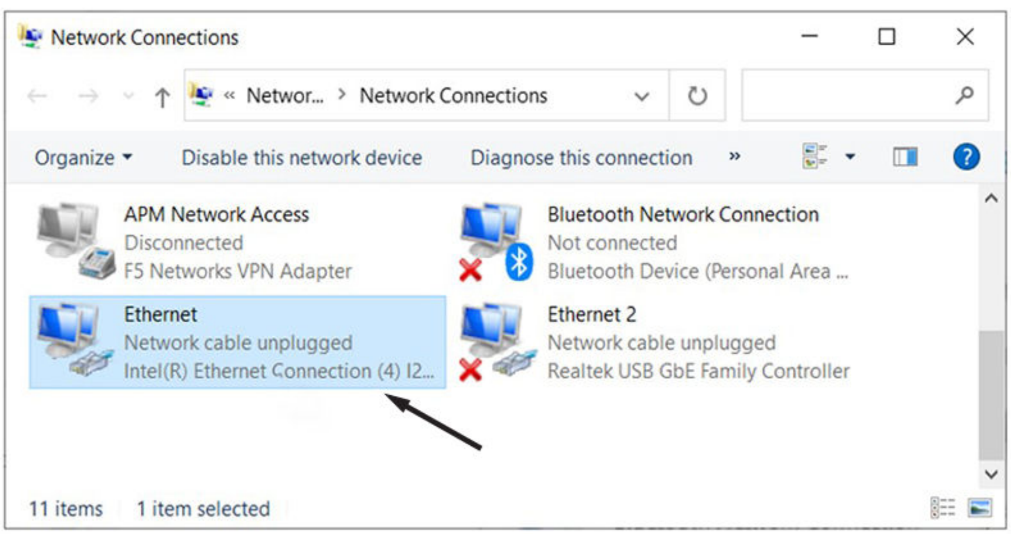
4. Click **Change adapter options**.

Figure 7-4 Network status - Change adapter options



5. Double-click **Ethernet**.

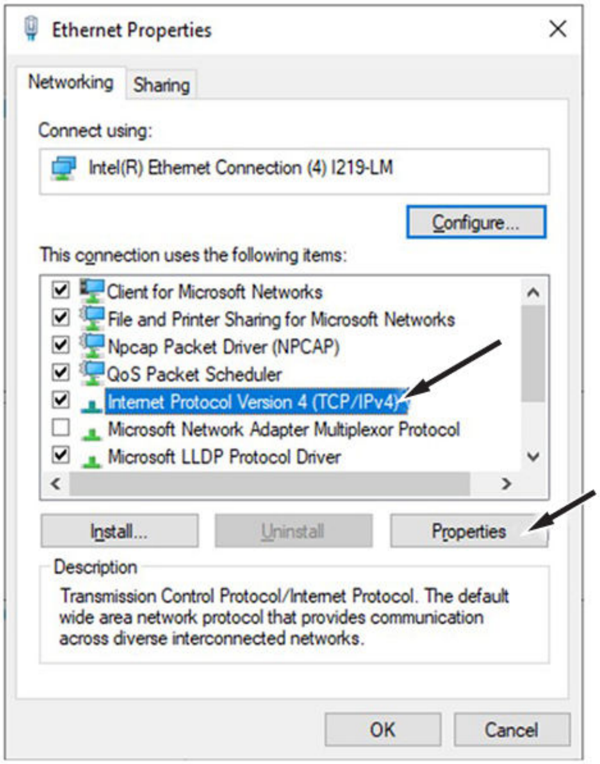
Figure 7-5 Network Connections - Ethernet option



6. Select **Internet Protocol Version 4 (TCP/IPv4)**.

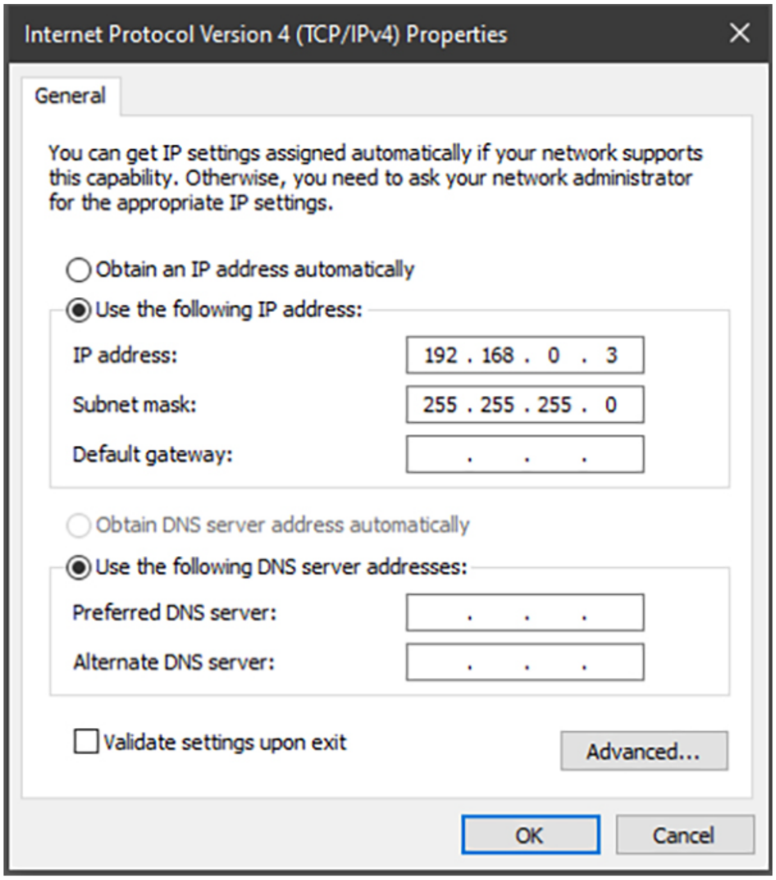
7. Click **Properties**.

Figure 7-6 Ethernet Properties



8. On the *Internet Protocol Version 4 (TCP/IPv4) Properties* window, select **Use the following IP address**.

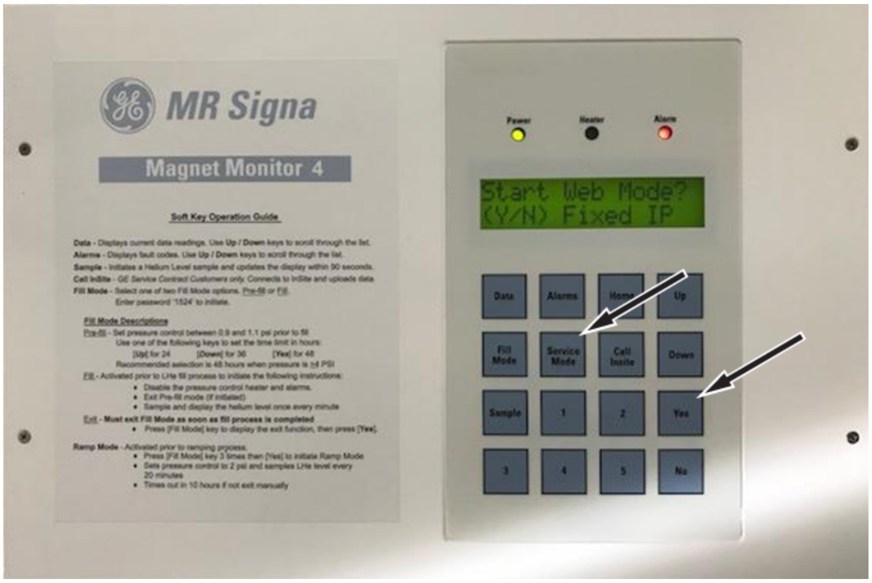
Figure 7-7 Use the following IP address option



9. Enter the following information:
   - **IP address**: `192.168.0.3`
   - **Subnet mask**: `255.255.255.0`
10. To save settings and exit, click **OK**.
11. To save the new local area connection properties, click **OK** or **Cancel**. This may take a few minutes.
12. Close the *Ethernet Properties* window.
13. Connect the Ethernet cable between the laptop and Magnet Monitor.
14. Activate the web interface:
    a. Press the **Service Mode** button on the Magnet Monitor's keypad until the LCD display shows **Start Web Mode? (Y/N) Fixed IP**.

b. Press **Yes**.

Figure 7-8 Activating the web interface



15. Log on to the web interface:

    a. Open an Internet browser window on the laptop/PC.

    b. Enter `192.168.0.1` in the address bar and press **Enter**.

    NOTE

    Some browsers may show a non-secure connection warning. This is normal and the connection will be confirmed.

    The browser should show the *Magmon4 Service Login* page.

16. Log on to the Magnet Monitor. See Chapter 8 Logging on to the Magnet Monitor as the customer on page 36.

8

Chapter 8 Logging on to the Magnet Monitor as the customer

Prerequisites

All Magnet Monitor units are pre-configured by the manufacturer in Non-Proprietary mode before the units ship to GE. Customers not on a GE service contract will need access to the unit to be able to do some configuration tasks in the event of a hardware failure where the Magnet Monitor would need replacement.

Required conditions
Laptop is directly connected to the Magnet Monitor using either a network cable or through the network.
Laptop communication is configured to support the appropriate network.
Laptop is communicating with the Magnet Monitor unit and the logon page shown on the laptop's web browser.

About this task

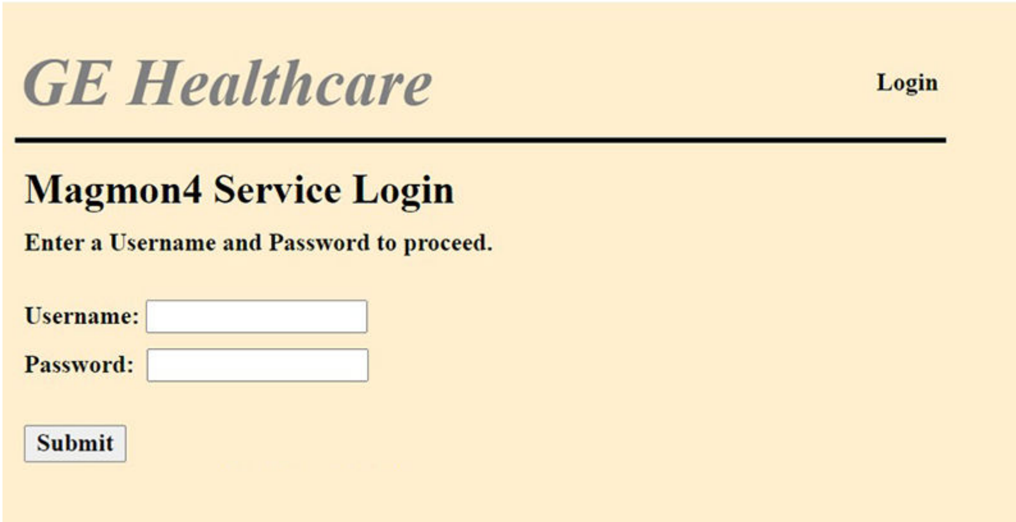
The unit can do these basic functions in Non-Proprietary mode:

- Show the last sample of He Level (%).
- Let you initiate the He level sample to update the information shown on the display.
- Show the He vessel pressure (psig).
- Do pressure control for magnets with recondenser refrigeration systems.
- Show alarms when a monitored parameter is out of range.
- Let you access Data mode for viewing current sensor values.
- Let you access Fill mode to increase He level sampling during the fill operation.

Procedure

1. On the *Magmon4 Service Login* page, fill the following fields (case-sensitive):
    - **Username**: `MMConfig`
    - **Password**: `ConfigMonitor`

Figure 8-1 Magmon4 Service Login window



NOTE

The Magnet Monitor will disconnect users after 45 minutes.

NOTE

IP 0.0.0.0 indicates a network connection did not occur.

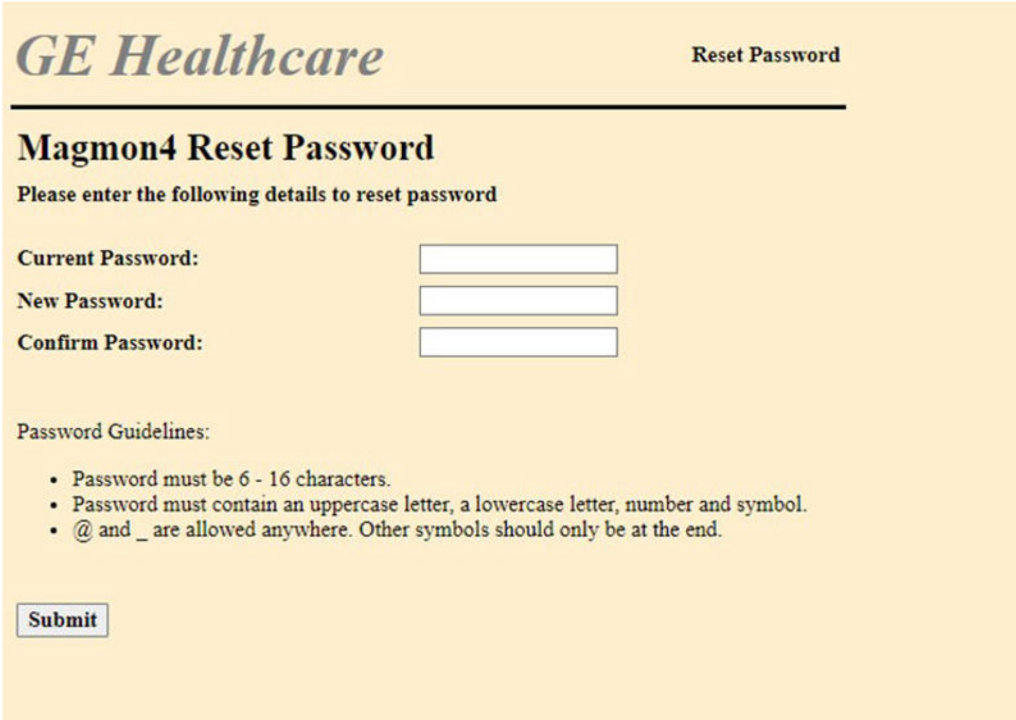
2. Click **Submit**.
3. After the password has been entered the first time for any user, a window will open requiring a password change. The password must be between 8 and 16 characters, and contain the following:

   NOTE

   Default passwords cannot be reused.

   - at least one lowercase letter
   - at least one uppercase letter
   - at least one digit
   - at least one symbol

Figure 8-2 Reset password



NOTE

Screen is not accurate. The password length must be 8 to 16 characters.

4. Click **Submit**. You will be logged out automatically.
5. Log on again with the new password to configure the Magnet Monitor.
6. Communicate the new password to the customer according to their site's Password Management Policy.

Chapter 9 Web page overview

9

9.1 Basic web page layout

Basic web page layout

The web pages and detailed information are provided in the order that they show in the navigation menu on the left side of the page.

This web page shows the basic layout of the Magnet Monitor web interface. The left column is a navigation menu where you can select one of the available web pages. Selecting one of the items in the navigation menu changes the page shown in the main section. Upon logon, the main section will show the *Version Info* page.

Figure 9-1 Basic web page layout

9.2 Configuration setup

9.2.1 Setting the date and time

About this task

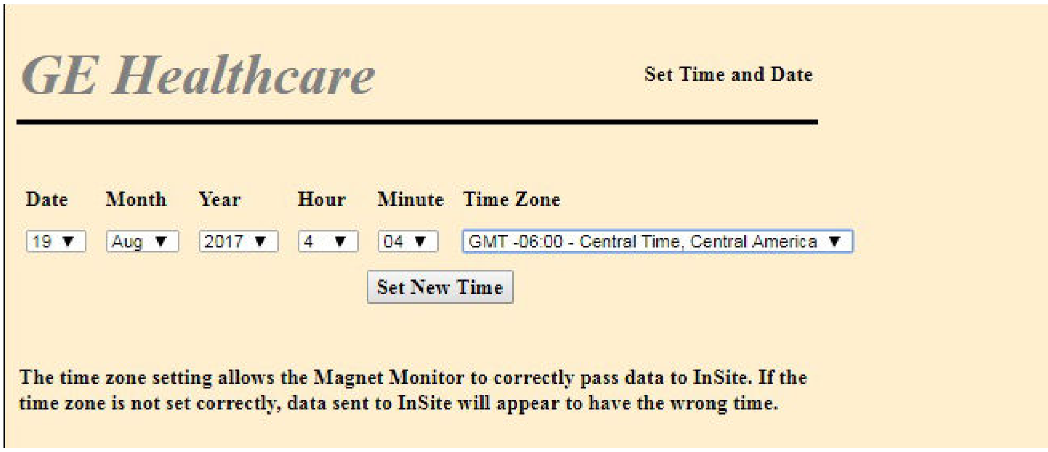
Use the *Set Time and Date* page to set the internal time and date for the Magnet Monitor. The internal time and date are used to timestamp the minute data and the event log items. The time zone setting is required to synchronize the Magnet Monitor's time to the back office time. If the time is changed forward, you may have to log back on if the 45 minutes is exceeded and the web server times out.

Procedure

1. On the *Set Time and Date* page, select a value from each list for the following fields:
   - **Date**

- **Month**
- **Year**
- **Hour**
- **Minute**

Figure 9-2 Set Time and Date page



2. Set the time zone by selecting a value for your area from the **Time Zone** menu. For example, if in Germany, select: `GMT +1`.
3. Click **Set New Time**.

9.2.2 Setting the magnet type

About this task

The *Magnet Type* page configures the Magnet Monitor to one of the predefined magnet types and compressor types. The number of sensors, type of compressors, and alarm parameters will vary between magnet model types, so the magnet type selection is a key element in the operation of the magnet and Magnet Monitor.

Procedure

1. The magnet serial number can be used to identify the magnet type for GE magnets. Locate the magnet serial number on the magnet's rating plate. The first letter (or letters) of that serial number indicates what type of magnet it is.
2. Select the magnet type, based on the serial number, from the list that most closely matches the system type that the Magnet Monitor will be monitoring. Other magnets should be selected based on the description in the Magnet Monitor. If you are unsure about what magnet type to select, contact the magnet support team or your regional Online Center.

NOTE

Each time the Magnet Type changes, the pressure set point will reset to the default value.

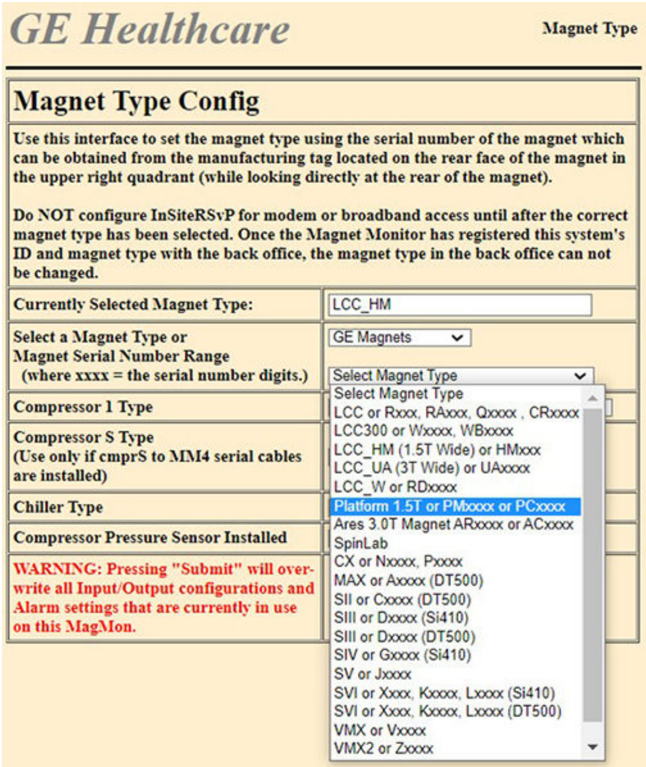
Table 9-1 Magnet types based on serial number

Serial number begins with	Select this magnet type
Nxxxx	CX

Table 9-1 Magnet types based on serial number (Table continued)

Serial number begins with	Select this magnet type
Pxxxx	CX
Rxxxx	LCC
RAxxxx	LCC
Qxxxx	LCC
CRxxxx	LCC
RDxxxx	LCC_W
HMxxxx	LCC_HM (1.5T wide)
Wxxxx	LCC300
WBxxxx	LCC300
UAxxxx	LCC_UA (3T wide)
PMxxxx	Platform 1.5T
PCxxxx	Platform 1.5T
ARxxxx	Ares 3.0T
ACxxxx	Ares 3.0T
Axxxx	MAX
Cxxxx	SII
Dxxxx	SIII
Gxxxx	SIV
Jxxxx	SV
Kxxxx	SVI
Lxxxx	SVI
Xxxxx	SVI
Vxxxx	VMX
Zxxxx	VMX2
7xxxx	7T_2CH

Figure 9-3 Magnet Type page



3. Select the proper compressor type, according to Table 9-2 Compressors on page 43, from the **Compressor 1 Type** menu.

   Compressor S Type should only be used when a Sumitomo F50SH compressor is connected with a serial cable (5807125) to the Magnet Monitor.

   Important

   Failure to configure the compressor type correctly will result in the Magnet Monitor being in constant alarm.

Figure 9-4 Compressor 1 Type menu

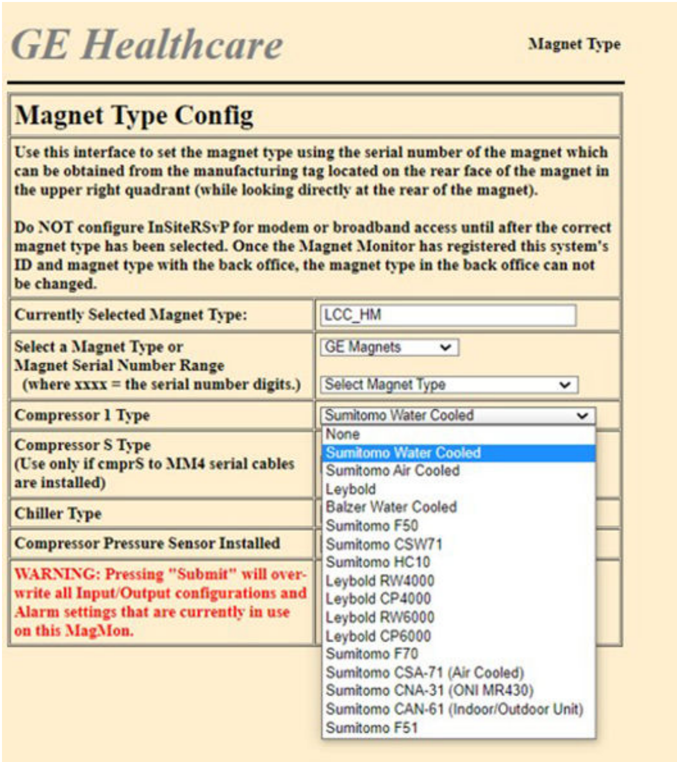
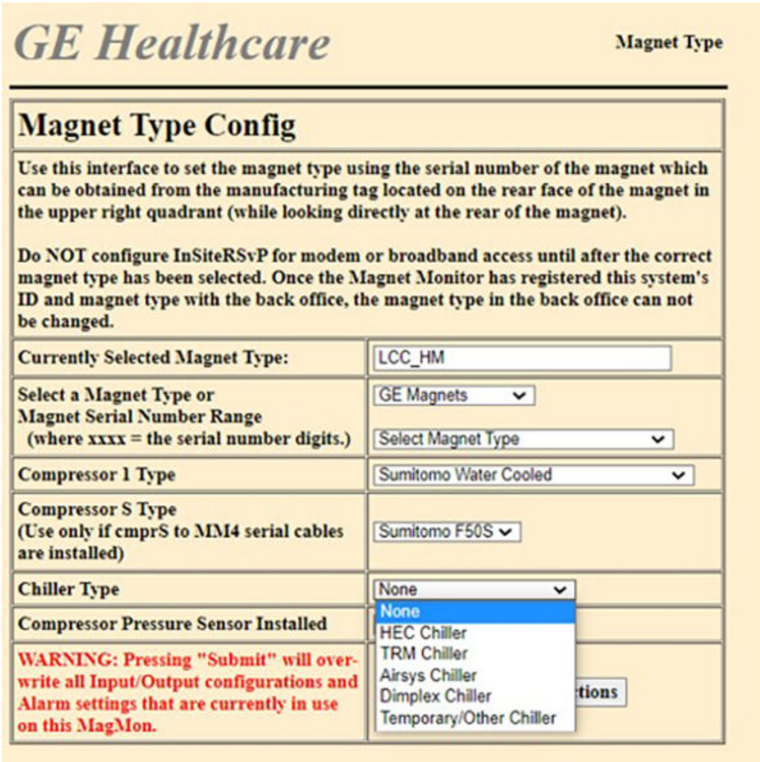


Table 9-2 Compressors

Compressor descriptions	
None	Sumitomo CSA-71 (Air Cooled)
Balzer Water Cooled	Sumitomo CSW71
Leybold	Sumitomo FA-50SH
Leybold CP4000	Sumitomo FA-50SL
Leybold CP6000	Sumitomo F50
Leybold RW4000	Sumitomo F50SH
Leybold RW6000	Sumitomo F51
Sumitomo CNA-31 (ONI MR430)	Sumitomo F70
Sumitomo CNA-61 (Indoor/Outdoor Unit)	Sumitomo HC10

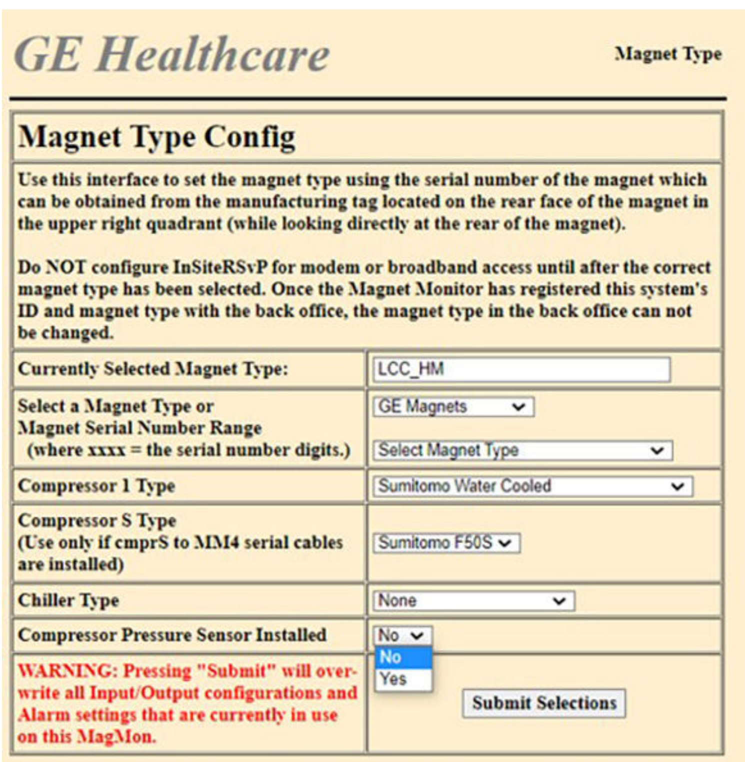
4. Select the suitable compressor cooling option from the **Chiller Type** menu.

Figure 9-5 Chiller Type menu



5. If an external compressor pressure sensor is installed, select **Yes** from the **Compressor Pressure Sensor Installed** menu. Otherwise select **No**.

Figure 9-6 Compressor Pressure Sensor Installed menu



6. Click **Submit Selections** after configuring the settings on the *Magnet Type* page.

9.2.3 Setting the network IP configuration

About this task

On this panel, the network information is configured to let the Magnet Monitor communicate with the back office.

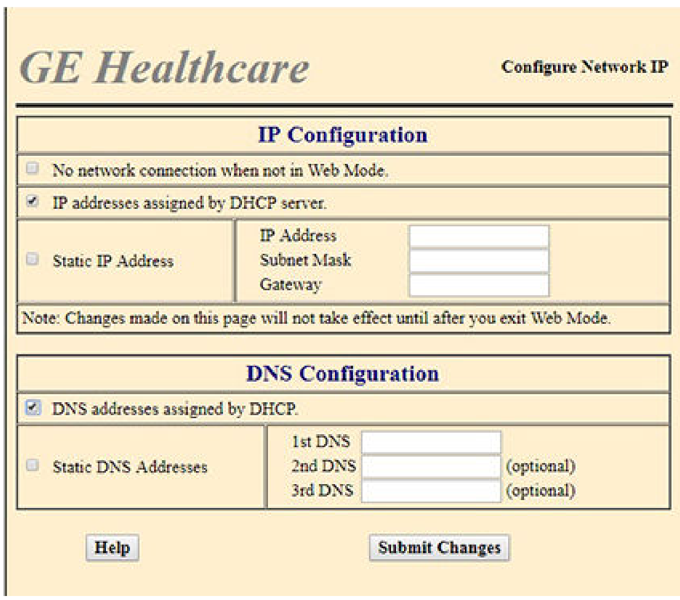
Procedure

1. If there is no communication requested by the customer, select the **No network connection when not in Web Mode** check box under the IP Configuration section. Otherwise, select **IP addresses assigned by DHCP server** or **Static IP Address** in the *IP Configuration* section, or **Static DNS Addresses** in the *DNS Configuration* section.
2. If **Static IP Address** is selected, enter the information provided by the customer's network administrator for IP Configuration and DNS Configuration.

   NOTE

   If **Static DNS Addresses** or **Static DNS Addresses** is selected, at a minimum the **1st DNS** field must be filled with the IP address.

   Figure 9-7 Configure Network IP page

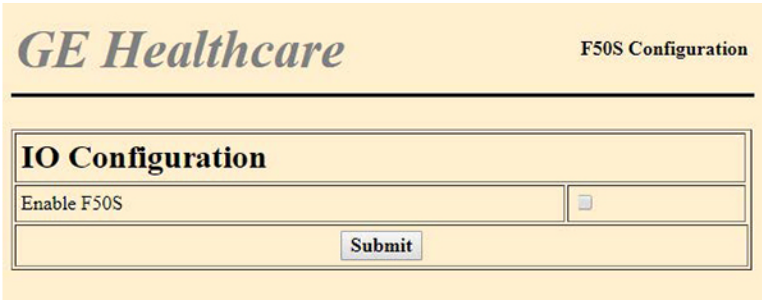
   

3. Click **Submit Changes**.

9.2.4 Configuring F50S

The *F50S Configuration* page shows the current setting of serial communication for the F50SH compressor. This interface lets the Magnet Monitor acquire additional information about the compressor. If the serial connection is made between the F50SH and the Magnet Monitor, select the **Enable F50S** check box to enable F50S and click **Submit**. The compressor type should also be selected on the *Magnet Type* page. The other parameters on the page should not need adjusting unless requested by the back office.

Figure 9-8 F50S Configuration page



Revision history

Revision	Date	Description
Controlled document for English is posted as DOC2587151.		
2	January 2023	Applied SIMS style to existing content. Updated for firmware version 4.3.0.
1	September 2019	Initial release.